

ATOMIC NUCLEUS

GLUON

Gluons hold quarks together.

Baryons (such as protons and neutrons) are made up of three quarks. A proton consists of two up quarks and one down quark.

PROTON

NEUTRON

Neutrons consist of two down quarks and one up quark and have a mass slightly higher than that of a proton.

SMALLER THAN AN ATOM

Atoms are built from elementary matter particles, such as quarks. There are six quark flavors: up, down, strange, charm, top, and bottom. Quarks also have a unique property known as “color charge” (see p.135), which is unrelated to color in the everyday sense. Particles with color charge cannot be isolated, so quarks always combine to form “colorless” composite particles such as protons, which are bound together through the strong force.